



# PYTHON BOOTCAMP

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# STRING MANIPULATION



## String Creation:

- Single Quote (one liner)
- Double Quote (one liner)
- Triple Quote (multi liner)

```
4 single_quote = 'Hello'
5 double_quote = "World"
6 triple_quote = """Multi-line
7 string"""
```

# STRING MANIPULATION



## String indexing and slicing:

- Indexing: number(position) for each character
- Slicing: grabbing multiple characters [start:stop:step]

```
10 text = "Python Programming"
11
12 print(text[0])      # (first character)
13 print(text[-1])     # (last character)
14 print(text[0:6])    # (slice 0 to 5)
15 print(text[:6])     # (from start to 5)
16 print(text[7:])     # (7 to end)
```

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## String Methods:

- len(): Length
- strip(): remove space
- upper(): UPPERCASE
- lower(): lowercase
- title(): Titlecase
- replace(): ("old","new")

```
19 name = " bob the builder "  
20  
21 print(len(name))           # Length  
22 print(name.strip())        # Remove whitespace  
23 print(name.upper())        # Uppercase  
24 print(name.lower())        # Lowercase  
25 print(name.title())        # Title case  
26 print(name.replace("bob", "jane")) # Replace
```

# STRING MANIPULATION



## String Formatting:

- f-string
- str.format()
- %-formatting

```
29 name = "John Doe"
30 age = 30
31
32 message_1 = f"My name is {name} and I am {age} years old." # f-strings
33 message_2 = "My name is {} and I am {} years old".format(name, age) # str.format()
34 message_3 = "My name is %s and I am %d years old." % (name, age) # %-formatting
35
36 print(message_1)
37 print(message_2)
38 print(message_3)
```

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## Exercise:

1. Build a simple text analyzer that counts words, characters, and sentences in a given text.

```
text = """Python is a powerful programming language. It's easy to learn  
and versatile!
```

```
You can use Python for web development, data science, and  
automation. The syntax is clean and readable.
```

```
This makes Python perfect for beginners and experts alike."""
```